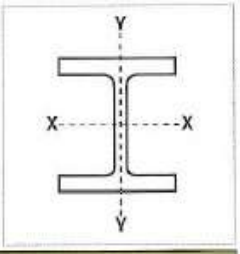


4b (i) Beams (Metric Units) - Part I

Designation		Height Of Beam	Width Of Beam	Thickness		Root Radius	Depth Between Fillets	Area Of Section	Ratios For Local Buckling		Dimensions For Detailing		
Size	Units Weight			Flange	Web				Flange	Web	End Clearance	Notch	
		H	B	T	t	r	d	A	b/T	d/t	C	N	n
mm	kg/m	mm	mm	mm	mm	mm	mm	cm^2			mm	mm	mm
450 x 200	65.1	446	199	12	8	13	396	83.0	8.29	49.5	6	106	25
	66.2	446	199	12	8	18	386	84.3	8.29	48.3	6	106	30
	74.9	450	200	14	9	13	396	95.4	7.14	44.0	7	106	27
	76.0	450	200	14	9	18	386	96.8	7.14	42.9	7	106	32
	88.9	456	201	17	10	18	386	113	5.91	38.6	7	106	35
	98.9	460	202	19	11	18	386	126	5.32	35.1	8	106	37
	110	460	205	19	14	18	386	140	5.39	27.6	9	106	37
450 x 300	106	434	299	15	10	24	356	135	9.97	35.6	7	155	39
	121	440	300	18	11	13	378	154	8.33	34.4	8	155	31
	124	440	300	18	11	24	356	157	8.33	32.4	8	155	42
	145	446	302	21	13	24	356	184	7.19	27.4	9	155	45
500 x 200	77.9	496	199	14	9	13	442	99	7.11	49.1	7	105	27
	79.5	496	199	14	9	20	428	101	7.11	47.6	7	105	34
	88.2	500	200	16	10	13	442	112	6.25	44.2	7	105	29
	89.6	500	200	16	10	20	428	114	6.25	42.8	7	105	36
	102	506	201	19	11	13	442	129	5.29	40.2	8	105	32
	103	506	201	19	11	20	428	131	5.29	38.9	8	105	39
	117	512	202	22	12	20	428	148	4.59	35.7	8	105	42
500 x 300	111	482	300	15	11	13	426	141	10.0	38.7	8	155	28
	114	482	300	15	11	26	400	146	10.0	36.4	8	155	41
	125	488	300	18	11	13	426	159	8.33	38.7	8	155	31
	128	488	300	18	11	26	400	164	8.33	36.4	8	155	44
	150	494	302	21	13	26	400	191	7.19	30.8	9	155	47
600 x 200	79.0	592	197	13	8	22	522	101	7.58	65.3	6	105	35
	92.5	596	199	15	10	13	540	118	6.63	54.0	7	105	28
	94.6	596	199	15	10	22	522	120	6.63	52.2	7	105	37
	103	600	200	17	11	13	540	132	5.88	49.1	8	105	30
	106	600	200	17	11	22	522	134	5.88	47.5	8	105	39
	118	606	201	20	12	13	540	150	5.03	45.0	8	105	33
	120	606	201	20	12	22	522	152	5.03	43.5	8	105	42
	134	612	202	23	13	22	522	171	4.39	40.2	9	105	45
600 x 300	133	582	300	17	12	13	522	169	8.82	43.5	8	154	30
	137	582	300	17	12	28	492	174	8.82	41.0	8	154	45
	147	588	300	20	12	13	522	187	7.50	43.5	8	154	33
	151	588	300	20	12	28	492	192	7.50	41.0	8	154	48
	170	594	302	23	14	13	522	217	6.57	37.3	9	154	36
	175	594	302	23	14	28	492	222	6.57	35.1	9	154	51
	203	602	304	27	16	28	492	259	5.63	30.8	10	154	55
	217	608	304	30	16	28	492	277	5.07	30.8	10	154	58
700 x 300	166	692	300	20	13	28	596	211	7.50	45.8	9	154	48
	182	700	300	24	13	18	616	232	6.25	47.4	9	154	42
	185	700	300	24	13	28	596	235	6.25	45.8	9	154	52
	215	708	302	28	15	28	596	274	5.39	39.7	10	154	56
800 x 300	191	792	300	22	14	28	692	243	6.82	49.4	9	153	50
	207	800	300	26	14	18	712	264	5.77	50.9	9	153	44
	210	800	300	26	14	28	692	267	5.77	49.4	9	153	54
	241	808	302	30	16	28	692	308	5.03	43.3	10	153	58
	267	816	303	34	17	28	692	340	4.46	40.7	11	153	62
900 x 300	210	890	299	23	15	18	808	267	6.50	53.9	10	152	41
	213	890	299	23	15	28	788	271	6.50	52.5	10	152	51
	240	900	300	28	16	18	808	306	5.36	50.5	10	152	46
	243	900	300	28	16	28	788	310	5.36	49.3	10	152	56
	283	912	302	34	18	18	808	360	4.44	44.9	11	152	52
	286	912	302	34	18	28	788	364	4.44	43.8	11	152	62
	304	918	303	37	19	18	808	387	4.09	42.5	12	152	55
	307	918	303	37	19	28	788	391	4.09	41.5	12	152	65



4b (i) Beams (Metric Units) - Part II

Surface Area Per Metre	Second Moment Of Area		Radius Of Gyration		Elastic Modulus		Plastic Modulus		Buckling Parameter	Torsional Index	Warping Constant	Torsional Constant	Designation	
	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	x-x	y-y	u	x	H	J	Unit Weight	Size
m^2	cm^4	cm^4	cm	cm	cm^3	cm^3	cm^3	cm^3			dm^6	cm^4	kg/m	mm
1.65	28130	1579	18.4	4.36	1262	159	1423	245	0.876	38.4	0.744	33.9	65.1	450 x 200
1.64	28700	1581	18.5	4.33	1287	159	1450	247	0.878	36.5	0.744	38.3	66.2	
1.66	32890	1870	18.6	4.43	1462	187	1652	290	0.878	33.6	0.889	51.6	74.9	
1.65	33450	1872	18.6	4.40	1487	187	1679	291	0.879	32.2	0.890	56.9	76.0	
1.67	40400	2308	18.9	4.51	1772	230	2003	356	0.884	27.5	1.11	92.5	88.9	
1.68	45430	2619	19.0	4.56	1975	259	2240	403	0.885	25.0	1.27	126	98.9	
1.68	47860	2742	18.5	4.43	2081	268	2399	423	0.869	23.9	1.33	152	110	
2.00	46800	6695	18.6	7.04	2157	448	2384	686	0.883	27.0	2.94	104	106	450 x 300
2.04	54730	8106	18.9	7.26	2488	540	2757	823	0.885	24.9	3.61	141	121	
2.02	56070	8114	18.9	7.18	2549	541	2825	828	0.887	23.4	3.61	163	124	
2.03	66380	9658	19.0	7.24	2977	640	3323	981	0.886	20.5	4.36	253	145	
1.75	40840	1843	20.3	4.31	1647	185	1869	288	0.874	37.5	1.07	52.5	77.9	500 x 200
1.74	41870	1846	20.3	4.27	1688	186	1914	290	0.876	35.2	1.07	60.8	79.5	
1.76	46810	2138	20.4	4.36	1872	214	2130	333	0.875	33.3	1.25	75.9	88.2	
1.75	47850	2142	20.5	4.33	1914	214	2175	335	0.877	31.6	1.25	85.9	89.6	
1.77	55480	2578	20.7	4.47	2193	257	2496	399	0.880	28.7	1.53	119	102	
1.76	56520	2582	20.7	4.43	2234	257	2541	401	0.881	27.5	1.53	132	103	
1.77	65450	3035	21.0	4.52	2557	300	2913	469	0.884	24.3	1.82	193	117	
2.12	58270	6756	20.3	6.92	2418	450	2696	690	0.877	32.4	3.68	93.9	111	500 x 300
2.10	60370	6768	20.4	6.82	2505	451	2791	695	0.880	29.4	3.69	118	114	
2.13	68860	8106	20.8	7.14	2822	540	3132	825	0.887	28.0	4.48	143	125	
2.11	70960	8118	20.8	7.05	2908	541	3228	830	0.889	25.9	4.48	172	128	
2.13	83810	9663	20.9	7.11	3393	640	3792	984	0.888	22.8	5.40	265	150	
1.92	58240	1665	24.1	4.07	1968	169	2239	265	0.870	45.2	1.40	53.0	79.0	600 x 200
1.95	66640	1976	23.8	4.10	2236	199	2576	312	0.861	42.9	1.67	69.2	92.5	
1.93	68720	1982	23.9	4.06	2306	199	2651	315	0.864	39.7	1.67	82.4	94.6	
1.96	75560	2274	24.0	4.16	2519	227	2904	358	0.863	38.4	1.93	97.3	103	
1.94	77630	2280	24.0	4.12	2588	228	2979	361	0.865	36.0	1.94	113	106	
1.97	88320	2716	24.3	4.26	2915	270	3357	426	0.868	33.4	2.33	147	118	
1.95	90400	2723	24.3	4.23	2983	271	3432	429	0.870	31.7	2.34	167	120	
1.97	103500	3178	24.6	4.32	3382	315	3893	498	0.873	28.3	2.76	237	134	
2.32	98950	7659	24.2	6.73	3400	511	3822	786	0.876	35.5	6.11	137	133	600 x 300
2.29	102700	7675	24.3	6.63	3530	512	3963	793	0.879	32.1	6.13	173	137	
2.33	114400	9009	24.7	6.94	3889	601	4348	921	0.886	31.2	7.27	199	147	
2.30	118100	9025	24.8	6.85	4018	602	4489	928	0.888	28.8	7.28	241	151	
2.35	133600	10570	24.8	6.98	4497	700	5057	1077	0.885	27.3	8.62	304	170	
2.32	137300	10590	24.9	6.90	4624	701	5197	1085	0.887	25.5	8.63	356	175	
2.34	162600	12680	25.1	7.00	5401	834	6101	1292	0.888	22.3	10.5	551	203	
2.35	179300	14090	25.4	7.13	5896	927	6653	1431	0.893	20.5	11.8	702	217	
2.51	172400	9030	28.6	6.53	4984	602	5629	936	0.878	34.3	10.2	260	166	700 x 300
2.54	197500	10820	29.2	6.83	5643	721	6338	1110	0.887	31.4	12.4	343	182	
2.53	201500	10830	29.3	6.78	5757	722	6464	1116	0.889	30.0	12.4	383	185	
2.55	237100	12890	29.4	6.86	6699	854	7559	1323	0.889	26.3	14.9	588	215	
2.71	253600	9936	32.3	6.39	6405	662	7288	1036	0.873	36.8	14.7	341	191	800 x 300
2.74	286400	11720	33.0	6.67	7159	781	8098	1210	0.882	33.9	17.6	440	207	
2.72	291700	11740	33.0	6.62	7292	782	8243	1216	0.883	32.5	17.6	486	210	
2.74	339200	13820	33.2	6.70	8397	915	9534	1426	0.883	28.7	20.9	726	241	
2.76	383600	15820	33.6	6.82	9402	1044	10680	1625	0.887	25.8	24.2	1003	267	
2.92	338500	10280	35.6	6.20	7608	687	8750	1079	0.863	42.2	19.3	361	210	900 x 300
2.90	345300	10290	35.7	6.16	7760	688	8913	1085	0.865	40.2	19.3	403	213	
2.94	404500	12630	36.4	6.43	8989	842	10290	1317	0.872	35.9	24.0	580	240	
2.92	411300	12650	36.4	6.39	9139	843	10450	1324	0.873	34.5	24.0	633	243	
2.97	491000	15660	36.9	6.59	10770	1037	12340	1622	0.876	30.1	30.2	979	283	
2.95	497800	15670	37.0	6.56	10920	1038	12500	1629	0.877	29.3	30.2	1050	286	
2.98	535400	17210	37.2	6.67	11660	1136	13380	1778	0.878	27.9	33.4	1234	304	
2.96	542200	17230	37.2	6.63	11810	1137	13540	1785	0.879	27.2	33.4	1316	307	